Aina Yohan RAKOTOARISON

FULL STACK DEVELOPER

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PROFILE

Passionate about software development and artificial intelligence, I am currently a full stack developer and a Master 2 MBDS student at Université Côte d'Azur. Holding a Bachelor's degree in Web and Design from IT University, I design and consume REST APIs, develop web applications with Angular/TypeScript and integrate AI tools. Rigorous and self-reliant, I am now looking to put my skills at the service of a team to take on new technical challenges.

PROFESSIONAL EXPERIENCE

Full Stack Developer

Aug 2025 — Present

Heaven Sound · Antananarivo

- Design and development of an integrated web platform (shop, donations, events, community) using agile methodology over 6 sprints: analysis, UML/data modeling (46 tables), 24 screens
- N-tier architecture: Angular 17, TypeScript and Tailwind CSS (front end), Node.js/TypeScript API, PostgreSQL and Firebase (Firestore)
- Online shop (catalog, cart, checkout, stock with movement history, Excel/PDF export) and back office with analytics dashboard
- Donations and crowdfunding: Vanilla Pay integration (Mvola, Orange Money, Airtel Money, Stripe, PayPal), secure webhooks, emailed receipts
- Events (unique QR codes, real-time scanning PWA), secure authentication (email PIN) and role-based access control
- In charge of the visual side · Online demo: https://heavensound.mg/

ERPNext Integrator

Feb 2025 — Jun 2025

IT University · Antananarivo

- Integration of features into ERPNext and of APIs connecting ERPNext to an external application.

MERN Full Stack Developer

Jan 2026 — Mar 2026

IT University (Academic project)

- Development of a shopping mall management web application with the MERN stack (MongoDB, Express.js, React and Node.js). Implementation of a REST API, secure authentication, and management of users, shops, products and orders.
- Online demo: https://www.smarket.qzz.io/

AI Developer (RAG)

Jul 2026 — Aug 2026

Personal project

- Design of a 100% local RAG system (“mini NotebookLM”) to query your own PDF/TXT/MD documents in natural language. Complete pipeline: extraction, chunking, multilingual embeddings, semantic search and generation of sourced answers. Python, LangChain, ChromaDB, Ollama, Streamlit.

Software Developer — Desktop & REST API

Aug 2026 — Sep 2026

Personal project · Antananarivo

- Windows client in Python/PyQt6 consuming a REST API: catalog, search, filters, pagination, seasons and episodes, libVLC player.
- Network call optimization (parallelization, caching), favorites, history and local SQLite cache.
- Packaging as a Windows executable with an automatic update system.

Mobile & Backend Developer — TATITRA Project

Sep 2026

Personal / academic project · Antananarivo

- Citizen reporting mobile application (Android/Kotlin, Jetpack Compose), offline-first MVVM architecture (Room, WorkManager)
- REST API (Node.js/Express), PostgreSQL database (Supabase) and web admin interface (React) with validation workflow
- Pair work, unit tests (JUnit, Node test) and Git management
- Source code: github.com/LoicLooney/Tatitra-app

C#/.NET Developer — Vehicle Inspection App

Oct 2026

Personal project · Antananarivo

- ASP.NET Core 10 REST API, SQL Server database and WPF (MVVM) desktop client; layered architecture, Entity Framework Core, DTOs.
- Inspection business rules (A/S/R results, re-inspection, due date, catalog of 76 defects).
- PDF inspection reports (QuestPDF), dashboard, JWT authentication with 3 roles.
- xUnit tests and GitHub Actions CI.
- Source code: github.com/Darkepsilon22/ControleTechnique

Mobile & Systems Developer — PhoneUnlock

Oct 2026

Personal project · Antananarivo

- Unlocking a Windows 11 session with a fingerprint from an Android phone, over the local network and with no cloud.
- Android app (Kotlin, Jetpack Compose, Keystore, BiometricPrompt) and C++20 Windows service with a Credential Provider (COM).
- Secure protocol: mutual TLS 1.3 with pinned keys, single-use requests signed with ECDSA P-256, encrypted password (DPAPI).
- 149 automated tests (102 C++, 47 Kotlin); installer with integrity checks and automatic rollback.
- Source code: github.com/Darkepsilon22/phone-unlock-windows

EDUCATION

Bachelor's Degree in Web and Design

2026

IT University

Master 1

2026

IT University

Master 2 MBDS

2026

Université Côte d'Azur

CERTIFICATIONS

Microsoft AI & ML Engineering Professional Certificate Oct 2026

SKILLS

- Programming languages: Java, Python, TypeScript, C#, Kotlin, C++
- Web application development (Angular/TypeScript)
- REST API design & integration (Java, Node.js, Python)
- Desktop software development (Python, PyQt6)
- .NET development (ASP.NET Core, Entity Framework Core, WPF)
- Android mobile development (Kotlin, Jetpack Compose)
- Relational databases (PostgreSQL, SQL Server, Oracle, MySQL)
- NoSQL databases (MongoDB, Firestore)
- Generative AI & LLMs (Prompt Engineering, AI tool integration)
- Systems modeling and design (UML, MVC)
- Git/GitHub

LANGUAGES

French — Fluent

English — Intermediate

Malagasy — Native

SOFT SKILLS

Problem solving · Autonomy · Rigor · Adaptability · Time management · Team spirit · Active listening · Perseverance · Creativity · Leadership

DRIVING LICENSE

B

ADDITIONAL INFORMATION

Interests: music and video games. Open to on-site or remote opportunities.